

safeguarding their liberties, travel and commercial activities, in a proactive manner. This requires vision, strategy and optimisation to enable selecting the most crucial technologies capable of providing maximum security for the resources invested. Utmost flexibility to address a wide range of threats is also required.

Aviation giant Boeing and other major aircraft manufacturers together are in process of developing a remote-control technology for flying a commercial aircraft. Had this technology been available in 2001, perhaps the 9/11 terror attacks on World Trade Center and Pentagon that killed nearly 3000 people could have been prevented.

Biometrics. There is research and development work going on for useful biometric options that include vein scans, facial thermographs, DNA matching, odour sensing, blood pulse measurements, skin-pattern recognition, nail-bed identification, gait recognition and ear-shape recognition.

The surest way to stop terrorists from hijacking or bombing planes from within is perhaps to stop attackers from boarding the planes in the first place. Hence, researchers have also envisioned combining sophisticated video surveillance at airports with software that analyses, for example, facial expressions for signs of lying. Researchers are also developing devices that can remotely scan for irregular blood pressure and heart rates, to serve as supplementary data.

Clothing. In the future, it is possible that flight marshals and crew would be adorning uniforms that also serve as body armour. A team of chemical engineers is researching the possibility of developing a thin but super-strong fabric by bonding 300 alternating layers of clay and polymer.

Some scientists in the USA, Europe and China have joined ordinary cotton with super-hard boron to create a boron nano-tube fabric that is



Fig. 7: Prof. Hany Farid, the father of robust hashing (Image courtesy: www.concordmonitor.com)



Fig. 8: Given the severe threat of ISIS across the world, there is need for various countries to have a common vision (Image courtesy: <http://abcnews.go.com>)

light and flexible enough to be put on as an undershirt, and could even shield the body from a bullet.

Robust hashing. Robust hashing is an innovative futuristic tool against online video of violence, beheadings and anti-Western criticism that are used to get new people into the terrorists' fold, and extend hate and violence.

These videos are being widely used by groups such as ISIS and *Taliban*. These are multiplying at lightning speed on social media sites. A software tool is being developed that creates an exclusive fingerprint for any video that can be used to automatically find copies as soon as these are uploaded, and this does not require human intervention or judgement.

Videos get edited or even altered. For example, an advertisement may be inserted making the video a little longer. However, the new technology beats that since it works by matching little snippets of the

video. Once a video has been tagged as related to terrorism by a viewer, a robust hash is created and positioned in a database, overseen by Counter Extremism Project. This database is used by various social media sites such as Facebook and Twitter so that they can automatically prevent uploading of this content.

While today several online and social media services are removing terrorism-inciting videos manually, it is hoped that in time to come it will be done automatically, making the whole process much more efficient and effective.

However, such software comes with its own problems since it can also automatically erase harmless videos such as a clip from a movie that has violence. Also, one man's terrorism could turn out to be another man's social protest.

Endpoint

The terrorist threat against the free world is global, serious, challenging and persisting. The means and technologies that are practical, affordable and can counter the serious threat of terrorism are required to be developed using joint efforts globally.

There are many impediments and challenges such as the tremendous expenditure involved, given the high cost of procuring and adopting a technology and scarcity of required resources, especially human skills as well as facilities and equipment. Any new technology may also create unintended consequences, bring unexpected liabilities, and produce environmental pollution and adverse public reactions.

Also, new technologies and systems will have significant training and leader development challenges. Obstacles to creating antiterrorism technologies that are steps ahead as compared to those used by terrorists are many. Hence, various countries need to create a vision and develop policies that will help overcome the barriers to innovation. **EFY**